

COMICCRAFT – PRESENTATION OVERVIEW

AI COMIC STORY CREATOR USING GEMINI MODELS

1. Project Title

ComicCraft – AI Comic Story Creator Using Gemini Models

2. Project Introduction

ComicCraft is a web-based Generative AI application that creates personalized comic stories from simple user-provided ideas.

The application uses AI to generate story content, comic panels, scene descriptions, dialogue, and image prompts.

The generated content is then combined into a complete comic and can be exported as a PDF.

3. Problem Statement

Creating a comic manually requires time and effort for:

- Story writing
- Character creation
- Scene development
- Dialogue writing

- Image creation
- Panel arrangement
- Final document preparation

ComicCraft aims to reduce this manual effort through AI-assisted comic creation.

4. Project Objectives

The main objectives are:

- Generate stories using Generative AI.
- Convert stories into structured comic panels.
- Generate visual content for panels.
- Create an organized comic layout.
- Export the completed comic as a PDF.
- Provide an easy-to-use web interface.

5. Proposed Solution

ComicCraft accepts a story idea from the user and processes it through multiple application modules.

The system uses Gemini AI to generate the story and panel information.

The image generation module creates visual content, while the comic layout and PDF modules prepare the final output.

6. Technology Stack

Frontend

- HTML
- CSS
- Jinja2 Templates

Backend

- Python
- FastAPI

AI

- Gemini Generative AI

Image Processing

- Pillow
- Image Generation Module

PDF

- FPDF / PDF Generation Library

Development

- Visual Studio Code
- GitHub
- Python Virtual Environment

7. System Architecture

The ComicCraft system consists of:

- User Interface
- FastAPI Backend
- Gemini AI
- Image Generation
- Comic Layout
- PDF Generation

Architecture

User

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Web Interface

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FastAPI Backend

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Gemini AI

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Comic Panels

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Image Generation

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Comic Layout

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PDF Generation

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Final Comic

8. Working Process

Step 1

The user enters a story idea.

Step 2

The FastAPI backend validates the input.

Step 3

Gemini AI generates the story outline and detailed comic panels.

Step 4

Image prompts are created for each panel.

Step 5

The image generation module processes the visual prompts.

Step 6

Text and images are combined into the comic layout.

Step 7

The final comic is converted into a PDF.

Step 8

The user can access the generated comic output.

9. Key Features

- AI story generation
- Structured comic panels
- Scene descriptions
- Character dialogue
- Narration and captions
- Image prompt generation
- Panel image generation
- Comic layout
- PDF export
- Input validation
- Error handling

10. Testing

The project includes testing of:

- Application launch
- Prompt input
- Input validation

- AI story generation
- Comic panel generation
- Image processing
- Comic layout
- PDF generation
- API communication
- Error handling

11. Expected Output

The expected output is a personalized AI-generated comic containing:

- Story panels
- Panel descriptions
- Captions
- Narration
- Dialogue
- Visual images
- Final PDF

12. Advantages

ComicCraft provides:

- Faster comic creation

- Reduced manual effort
- AI-assisted storytelling
- Automated panel generation
- Integrated image generation
- Easy PDF export
- Simple user workflow

13. Future Enhancements

Future improvements may include:

- Multiple art styles
- Character consistency
- Multilingual support
- Voice-based story input
- User-selected characters
- Cloud deployment
- User accounts
- Online comic sharing
- Additional AI models

14. Conclusion

ComicCraft demonstrates the use of Generative AI in creative content generation.

By combining Gemini AI, image generation, web technologies, comic layout, and PDF generation, the project provides an end-to-end workflow for creating personalized digital comics.

The project can be further enhanced with additional AI capabilities, visual styles, and cloud-based features.